Who Gets the Power? (Bijli Kis Ko?)

Fair load-shedding with quantum optimization

Qiskit Fall Fest 2026

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Repository: https://github.com/wajih-rathore/who-gets-the-power

1. Problem

In a heatwave there is not enough electricity for everyone, so utilities in Pakistan switch areas off in turns. Today they choose by "losses": the gap between power sent to an area and power paid for. An area with high losses is cut the longest, so people there who pay their bills sit in the dark too.

- Express Tribune, 2 Sep 2026: NEPRA ordered K-Electric to end this practice. Very-high-loss feeders averaged 12 h 14 min of outages a day, which the regulator called "collective punishment". (https://tribune.com.pk/story/2627009/nepra-orders-k-electric-to-end-load-shedding)
- Dawn, 26 May 2025: 2,000 MW of electricity was announced for Bitcoin mining and AI data centres. It was announced; we do not claim it is running. (https://www.dawn.com/news/1913342)

Our demo case is an Islamabad-style circuit with four switchable blocks and one protected load: I-10 (homes, high loss), G-9 (homes), Blue Area (offices & shops), an I-9 AI data centre that can pause, and PIMS hospital, which is always on. The code's results file labels the four blocks N1, N2, N3 and DC (its example names are Karachi areas); the mapping is N1 = I-10, N2 = G-9, N3 = Blue Area, DC = I-9 AI centre. All area numbers are illustrative, not real utility data.

Block	Area	Illustrative loss %	Illustrative value
N1	I-10 (homes, high loss)	38	damage/hour: 1.6
N2	G-9 (homes)	22	damage/hour: 1
N3	Blue Area (offices & shops)	12	damage/hour: 3
DC	I-9 AI data centre (can pause)	-	revenue/hour: 5.5, 7, 4.5

Each block is one 10 MW chunk. revenue and damage in synthetic 'PKR million per hour'.

2. Formulation

Decision: x[b,t] = 1 if block b is switched off in hour t. 4 blocks x 3 hours (2–3 pm, 3–4 pm, 4–5 pm) = 12 binary variables. The hospital is not a variable; it is always on.

Rule: exactly k_t blocks must be off in hour t, with k = (1, 2, 1) on a normal afternoon. The rule is added as a quadratic penalty, so any plan that breaks it scores badly.

$$\text{minimize } w_f \Sigma_n (\Sigma_t x[n,t])^2 + w_c \Sigma_n \Sigma_t x[n,t]\cdot x[n,t+1] + w_r \Sigma_t r_t \cdot x[DC,t] + w_e \Sigma_{n,t} d_n \cdot x[n,t] + P \Sigma_t (\Sigma_b x[b,t] - k_t)^2,\ x \in \{0,1\}^{12}$$

In plain words, the plan should:

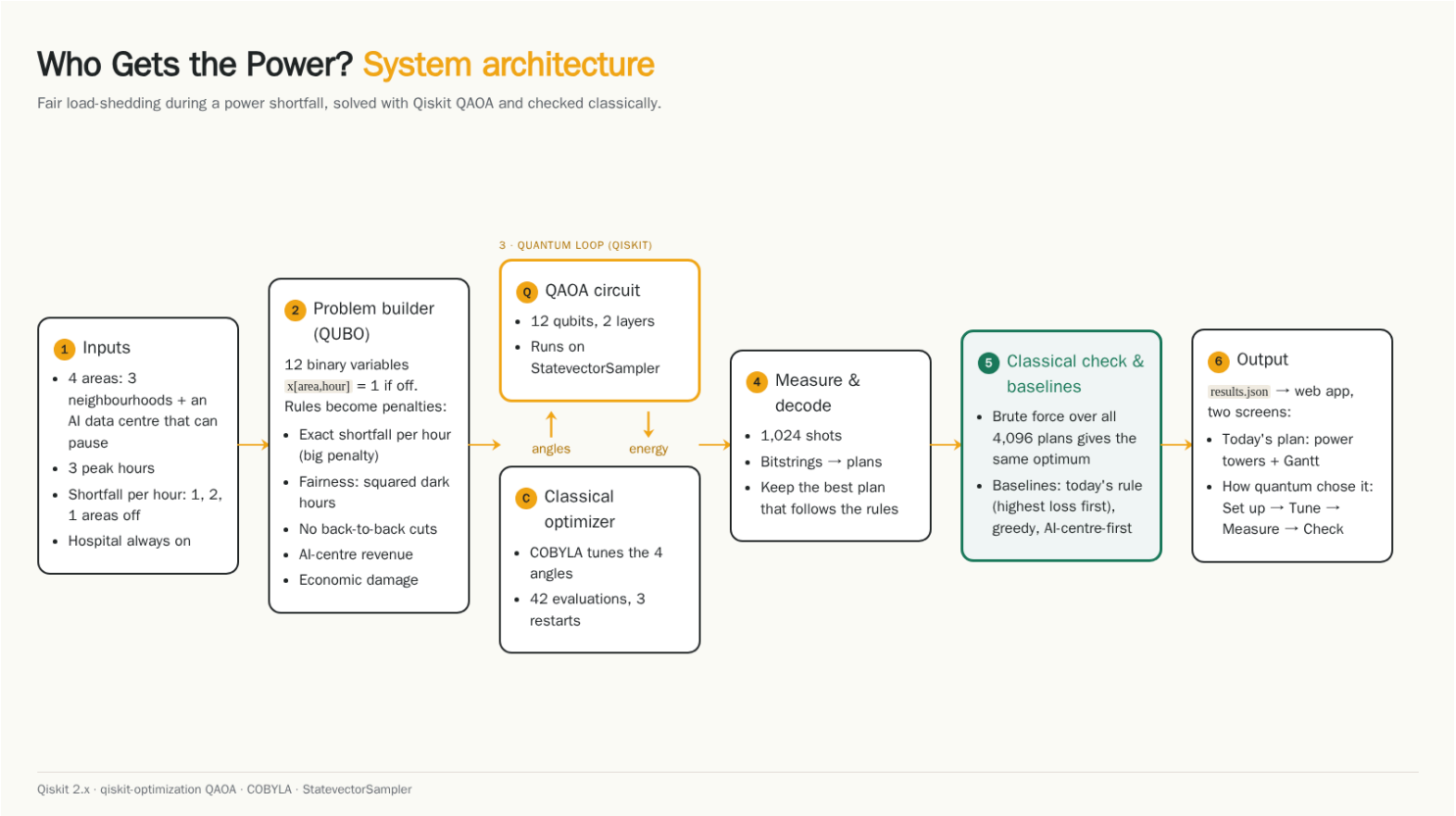
- Fairness: spread dark hours evenly (squaring punishes one area getting many hours).
- Back-to-back: avoid leaving an area dark two hours in a row.
- AI-centre revenue: count the revenue lost when the data centre pauses (r_t per hour).
- Economic damage: count the cost of cutting an area (d_n, higher for offices & shops).
- Exact shortfall: switch off exactly k_t blocks each hour (penalty P).

Weights (hand-tuned): fairness 2, back-to-back 3, revenue 1, damage 1, penalty P = 40.

3. Quantum method

- QAOA through qiskit-optimization, reps = 2 (4 angles), on 12 qubits.
- Classical optimizer: COBYLA, 3 restarts; the recorded trace has 42 evaluations (normal) and 45 (disaster).
- Simulator: Qiskit StatevectorSampler, 1024 shots, fixed seeds.
- Why QAOA: every decision is on/off and every rule is a pair term or a quadratic penalty, so the problem maps directly onto a QAOA cost Hamiltonian with no extra encoding.
- Grover Adaptive Search was tried on a 9-variable version: it needed 16 qubits, took 71 s and returned an infeasible answer, so we rejected it.

4. Architecture



- Inputs: illustrative area data (loss, damage, AI-centre revenue) and the hourly shortfall for each scenario.
- QUBO builder: turns the four goals and the exact-shortfall rule into one 12-variable quadratic model.
- Solvers: QAOA on the Qiskit simulator, brute force over all 4,096 plans as the exact reference, and three classical baselines (today's rule, data centre first, greedy).
- Output: results.json and charts, which feed the web demo and this report.

5. Results

Lower objective is better. Approx. ratio = exact objective / plan objective (1.0 is optimal). Max dark hours = the most hours any one area is off.

Normal hot afternoon: k = (1, 2, 1)

Power is short by one block at 2 pm, two at 3 pm and one at 4 pm.

Plan	Objective	Approx. ratio	Max dark hours	Who is off
Today's rule (cut highest losses)	33.16	0.52	3	I-10: 2–3 pm, 3–4 pm, 4–5 pm; G-9: 3–4 pm
Data centre first	21.24	0.81	1	I-10: 3–4 pm; I-9 AI centre: 2–3 pm, 3–4 pm, 4–5 pm

Greedy	31.84	0.54	3	I-10: 3–4 pm; G-9: 2–3 pm, 3–4 pm, 4–5 pm
QAOA	17.14	1	1	I-10: 3–4 pm; G-9: 3–4 pm; Blue Area: 2–3 pm; I-9 AI centre: 4–5 pm
Exact (brute force)	17.14	1	1	I-10: 3–4 pm; G-9: 3–4 pm; Blue Area: 2–3 pm; I-9 AI centre: 4–5 pm

QAOA statistic	Value
P(optimum): QAOA vs random	0.00566 vs 0.000244
P(valid plan): QAOA vs random	0.4587 vs 0.0234
QAOA run time	27 s
QAOA found the optimum	yes
Exact check	4096 plans (96 valid) in 0.035 s; agrees with QAOA: yes

Disaster: a power plant trips: k = (1, 3, 1)

A plant trips at 3 pm, so three blocks must go dark that hour instead of two.

Plan	Objective	Approx. ratio	Max dark hours	Who is off
Today's rule (cut highest losses)	39.36	0.61	3	I-10: 2–3 pm, 3–4 pm, 4–5 pm; G-9: 3–4 pm; Blue Area: 3–4 pm
Data centre first	24.64	0.97	1	I-10: 3–4 pm; G-9: 3–4 pm; I-9 AI centre: 2–3 pm, 3–4 pm, 4–5 pm
Greedy	38.04	0.63	3	I-10: 3–4 pm; G-9: 2–3 pm, 3–4 pm, 4–5 pm; Blue Area: 3–4 pm

QAOA	23.84	1	1	I-10: 3–4 pm; G-9: 3–4 pm; Blue Area: 3–4 pm; I-9 AI centre: 2–3 pm, 4–5 pm
Exact (brute force)	23.84	1	1	I-10: 3–4 pm; G-9: 3–4 pm; Blue Area: 3–4 pm; I-9 AI centre: 2–3 pm, 4–5 pm

QAOA statistic	Value
P(optimum): QAOA vs random	0.00601 vs 0.000244
P(valid plan): QAOA vs random	0.3473 vs 0.0156
QAOA run time	25.3 s
QAOA found the optimum	yes
Exact check	4096 plans (64 valid) in 0.035 s; agrees with QAOA: yes

Takeaways

- QAOA returned the exact optimum in both scenarios (approx. ratio 1.0, confirmed by brute force).
- Today's rule leaves one area (I-10) dark for 3 hours in a row; the fair plan's maximum is 1 hour.
- Today's rule scores about 1.9x worse than the optimum (33.16 vs 17.14).
- When a plant trips at 3 pm, the planner re-plans the afternoon and still keeps every area to at most 1 dark hour.

6. Classical comparison and honesty

At 12 variables brute force is faster: it checks all 4096 plans in 0.035 s, against about 27 s for QAOA on the simulator. We claim no quantum speedup. The value is the pipeline and a method that scales: the number of possible plans doubles with every added variable, so brute force stops being an option as areas and hours grow. Running on quantum hardware is future work.

7. Limitations

- Synthetic, illustrative data, not real feeder data.
- Equal-size blocks (10 MW each), a simplification of real feeders.
- Simulator only, not quantum hardware; no noise studied.

- Tiny instance (12 variables).
- Penalty and objective weights are hand-tuned.
- Only part of the announced 2,000 MW plan was implemented; the AI centre here is a scenario, not a confirmed facility.

8. Extensions and future work

- Done: plant-trip re-planning (the disaster scenario above).
- Done: trade-off sweep "protect homes ↔ protect AI revenue" (table below; each row is the exact optimum at that weight).
- Next: an XY (rule-preserving) mixer so QAOA only explores valid plans.
- Next: more areas and hours; real feeder data; runs on IBM quantum hardware.

Setting	Revenue weight	Who is off	Max dark hours	AI revenue kept	QAOA found optimum
Protect homes	0.5	G-9: 3–4 pm; I-9 AI centre: 2–3 pm, 3–4 pm, 4–5 pm	1	0%	no
Lean homes	0.8	I-10: 3–4 pm; G-9: 3–4 pm; I-9 AI centre: 2–3 pm, 4–5 pm	1	41.2%	yes
Balanced	1	I-10: 3–4 pm; G-9: 3–4 pm; Blue Area: 2–3 pm; I-9 AI centre: 4–5 pm	1	73.5%	yes
Protect revenue	2	I-10: 3–4 pm; G-9: 2–3 pm, 4–5 pm; Blue Area: 3–4 pm	2	100%	yes

9. Reproducibility

Repository: https://github.com/wajih-rathore/who-gets-the-power. From the repository root (Python 3.13, qiskit 2.6.2, qiskit-optimization 0.7.0, pinned in requirements.txt):

- python3 -m venv .venv && . .venv/bin/activate && pip install -r requirements.txt
- PYTHONPATH=src python -m bijli.run # writes web/data/results.json and results/*.png, about 2-3 minutes
- PYTHONPATH=src pytest -q # tests
- python report/make_diagram.py && python report/build_report.py # architecture diagram + .docx report
- python -m http.server -d web 8000 # demo, then open http://localhost:8000

10. References

- Express Tribune, 2 Sep 2026. NEPRA orders K-Electric to end load-shedding. https://tribune.com.pk/story/2627009/nepra-orders-k-electric-to-end-load-shedding
- Dawn, 26 May 2025. 2,000 MW announced for Bitcoin mining and AI data centres. https://www.dawn.com/news/1913342